PSEUDO-CATENARITY PASSES ALONG LOCALLY FINITE TYPE MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-geometry.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for stacks-geometry.tex (ge-occupied.json, 2026-10-14); 207 route=geometry / chapter-list field hits in focused-seeds are alias noise (route name and alphabetical chapter enumerations), excluded per the ladder alias-blind-spot caveat.

Source reference. Original reference 02J9: sources/stacks-chow/tags.txt, source lines 3328–3328. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *If $\mathcal{X}$ is a pseudo-catenary locally Noetherian algebraic stack, and if $\mathcal{Y} \to \mathcal{X}$ is a locally of finite type morphism, then there exists a smooth surjective morphism $V \to \mathcal{Y}$ whose source is a universally catenary scheme; thus $\mathcal{Y}$ is again pseudo-catenary.*

Proof. By assumption we may find a smooth surjective morphism $U \to \mathcal{X}$ whose source is a universally catenary scheme. The base-change $U \times_{\mathcal{X}} \mathcal{Y}$ is then an algebraic stack; let $V \to U \times_{\mathcal{X}} \mathcal{Y}$ be a smooth surjective morphism whose source is a scheme. The composite $V \to U \times_{\mathcal{X}} \mathcal{Y} \to \mathcal{Y}$ is then smooth and surjective (being a composite of smooth and surjective morphisms), while the morphism $V \to U \times_{\mathcal{X}} \mathcal{Y} \to U$ is locally of finite type (being a composite of morphisms that are locally finite type). Since U is universally catenary, we see that V is universally catenary (by Morphisms, Lemma 02J9), as claimed. □

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